Present-epoch scale matching, the weak-field invariant $C_{\rm obs}$, and constraints on a cubic T^3 geometric construction

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The coincidence $2\pi a_0 \sim cH_0$ is a standard observation in the MOND literature. We test one proposed route from that coincidence to a *geometric derivation*: (i) a compact flat cubic three-torus of present-day side $L = c/H_0$ with a putative "circulation quantum" $\kappa = cL$, and (ii) identification of the isotropic projector ratio $\mathrm{Tr}(h)/\mathrm{Tr}(\gamma) = 2/3$ with the weak-field coefficient in an AQUAL-type action. The route is assessed against explicit microphysical, cosmic-scale, and weak-field criteria. Writing $\kappa = cL$ does not by itself quantize circulation; the package $L = c/H_0$, $\Gamma = cL$, and $\omega = H_0$ is kinematically inconsistent if L is a circumference; and cubic E_1 topology bounds require $L \gtrsim \mathcal{O}(6)\, c/H_0$, which is incompatible with the $\alpha = 1$ scale match. Exact tracking $L_{\rm phys}(t) = c/H(t)$ is also incompatible with fixed comoving moduli in ordinary FLRW except for a coasting expansion. The projector identity is correct but topology-independent and absent from the weak-field action, so it does not fix $C_{\rm obs}$. We instead derive the normalization-invariant response combination $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$. In the static spherical baseline, imposing both $a_0 = cH_0/2\pi$ and the un-derived comparator $C_{\rm obs} = 2/3$ gives $a_{0,\rm eff} = (4/9)a_0$, corresponding to an approximately 18% shift in asymptotic baryonic Tully–Fisher velocity normalization. This is a conditional consistency check, not a global T^3 prediction or a completed galaxy-data fit. The retained results are a present-epoch phenomenological encoding of the coincidence, a clean EFT invariant, and a reconstruction checklist for any future geometric claim.

Keywords: Modified Newtonian dynamics; cosmic topology; effective field theory; radial acceleration relation; compact manifolds

I. INTRODUCTION

Empirical galactic dynamics exhibit a characteristic acceleration $a_0 \sim 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$ that organizes the radial acceleration relation (RAR) and the baryonic Tully–Fisher relation [1–4]. The numerical coincidence

$$2\pi\, a_0 \sim c\, H_0 \tag{1}$$

is well known [1, 2]. It is a legitimate research question whether a compact flat three-torus $T^3 = S^1 \times S^1 \times S^1$ [5, 6] and a weak-field effective field theory (EFT) can turn that coincidence into a derivation.

The distinction matters beyond topology bookkeeping. The normalization that enters the deep-MOND response also enters the RAR and the asymptotic baryonic Tully–Fisher relation, so a geometric coefficient must survive both action-level matching and observational normalization. Conversely, failure of the particular construction tested here would not decide the viability of MOND as a class, relativistic MOND theories, or other microscopic completions. It would show only that the empirical coincidence and the cubic topology do not supply the claimed derivation without additional physics.

The two-step construction examined in this note answers "yes":

(A) "Circulation quantization" on a domain of size $L = c/H_0$ yields $a_0 = cH_0/2\pi$;

(B) the transverse-projector ratio 2/3 yields the EFT coefficient $C_{\rm obs}$ in a deep-MOND square-root law.

Our analysis identifies mathematical incompatibilities in both steps under the stated assumptions. This short technical note records those scoped constraints and isolates what *does* follow cleanly from a static AQUAL-type action [7].

a. Evaluation framework. The analysis uses three groups of criteria, previewed here so that each later calculation has an explicit standard: (M1)–(M4) require a phase-bearing order parameter, integer winding, a consistent meaning of L, and a dynamical map to a_0; (S1)–(S2) require a viable modulus history and consistency with compact-topology bounds; and (E1)–(E3) require action-level projector dynamics, the correct RAR/BTFR normalization, and solved compensated periodic sources. The detailed, forward-looking checklist appears in Sec. VIII. Table I summarizes the resulting analytical dispositions.

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TABLE I. Analytical disposition of claims discussed in this work.

Quantity / claim	**Disposition**	**Note**
Empirical coincidence $2\pi a_0 \sim cH_0$	Fact	Motivating observation only
$a_0 \equiv cH_0/2\pi$ at the present epoch	Phen.	Named postulate; not derived
"Circulation quantization $\Rightarrow a_0$"	Not derived	$\kappa = cL$ is not h/m quantization
Package $(L, \Gamma = cL, \omega = H_0)$ with L a circumference	Incompatible	Inconsistent if L is a circumference
Cubic E_1 with $L = c/H_0$ vs. Planck topology	Incompatible	$L \gtrsim \mathcal{O}(6)\, c/H_0$ required
$L_{\rm phys}(t) = c/H(t)$ with fixed comoving moduli	Incompatible	Needs evolving moduli or coasting FLRW
Isotropic count $\mathrm{Tr}(h)/\mathrm{Tr}(\gamma) = 2/3$	Derived	Generic 3D identity; not special to T^3
$C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$	Derived	Field-normalization invariant
$C_{\rm obs} = 2/3$ from the trace ratio	Not derived	Projector absent from the action
Simultaneous $a_0 = cH_0/2\pi$ and $C_{\rm obs} = 2/3$	Rescaled	Conditional spherical normalization: $a_{0,\rm eff} = (4/9)a_0$; no data fit
Compensated sources on compact T^3	Derived	$\int \mathrm{div} = 0$
Geometric derivation of a_0 or $C_{\rm obs}$	Open	Path in Sec. VIII

II. CUBIC THREE-TORUS GEOMETRY AND NOTATION

A flat three-torus is the quotient $\mathbb{R}^3/\Lambda$ by a lattice of translations [5]. The correct schematic is a parallelepiped with opposite faces identified (Fig. 1), not the doughnut surface of revolution in $\mathbb{R}^3$. That doughnut is an embedding of T^2.

"T^3" alone does not imply equal periods. Isotropic statements require a *cubic* (or explicitly isotropised) lattice. Rectangular and twisted flat manifolds (E_2, E_3, ...) are different geometries with different CMB constraints [6, 8].

Throughout, L means a *physical* edge length at the epoch under discussion unless "comoving" is written explicitly. When a numerical Hubble scale is needed we take $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for illustration; none of the scoped arguments depends on that choice at the factor-of-two level.

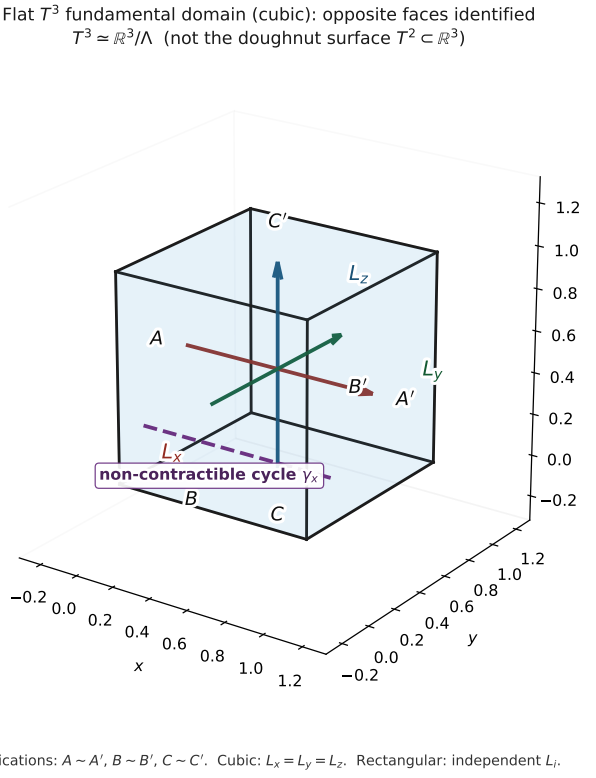


FIG. 1. Flat cubic T^3 fundamental domain: opposite faces identified ($A \sim A'$, $B \sim B'$, $C \sim C'$). Edge lengths L_x, L_y, L_z are the three independent periods. Cubic means $L_x = L_y = L_z$. This is not the embedded doughnut T^2.

III. $a_0 = cH_0/2\pi$ IS NOT CIRCULATION QUANTIZATION

A. What quantization requires

In a superfluid with order parameter $\Psi = \sqrt{n}\, e^{i\theta}$, the velocity follows from the phase gradient and

$$\Gamma = \oint \mathbf{v} \cdot d\mathbf{l} = n\,\frac{h}{m}\,, \qquad n \in \mathbb{Z}\,, \tag{2}$$

with single-valuedness of Ψ on closed cycles [9–11]. Equation (2) quantizes *phase winding*, not "length times a speed limit."

B. What the geometric proposals actually write

Those proposals introduce

$$\kappa_{\rm cosmo} \stackrel{?}{=} c\,L \tag{3}$$

by multiplying a maximum speed by a length. For constant speed on a cycle of length L, one has only $\Gamma = vL$; setting $v = c$ yields $\Gamma = cL$. No integer winding, no order parameter, and no mass m enter. That is a kinematic assignment, not quantization.

If one *did* set the $n = 1$ quantum equal to c^2/H_0,

$$m = \frac{hH_0}{c^2} \sim 10^{-32}\,\mathrm{eV}/c^2\,, \tag{4}$$

the microscopic mass reappears as a sharp prediction—the opposite of claiming that no microphysics is needed.

C. Internal inconsistency of the (L, Γ, ω) package

Suppose $L = c/H_0$ is the physical circumference of a closed cycle, so $R = L/(2\pi)$. Then:

- $\Gamma = cL$ implies $v = c$;
- $\omega = H_0$ with $v = \omega R$ implies $v = H_0 L/(2\pi) = c/(2\pi)$, contradicting $v = c$;
- the centripetal acceleration c^2/R is $2\pi c H_0$, not $cH_0/2\pi$.

Thus one cannot simultaneously maintain

$$L = \frac{c}{H_0}\,, \qquad \Gamma = cL\,, \qquad \omega = H_0 \tag{5}$$

under that geometric reading of L. "Distributing cH_0 across a 2π angular topology" is not coordinate-invariant: the range of an angular chart can be rescaled. A physical 2π arises only from a defined phase convention coupled to velocity.

a. Conclusion.

$$a_0 \equiv \frac{cH_0}{2\pi} \tag{6}$$

may be retained only as a *present-epoch phenomenological scale-matching postulate* that encodes Eq. (1). For $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ one finds $a_0 \approx 1.1 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, of the observed order of magnitude [3]. Under the assumptions tested here, "geometric derivation from circulation quantization" is unsupported unless the microphysical requirements (M1)–(M4) are supplied.

IV. CUBIC TOPOLOGY BOUNDS AND $L = c/H_0$

Planck topology analyses constrain cubic E_1 three-tori tightly. In terms of the radius R_i of the largest sphere inscribed in the fundamental domain, Planck reports $R_i > 0.97\,\chi_{\mathrm{rec}}$ for the cubic case at high confidence [6]. For a cube of side L, $R_i = L/2$, so

$$L \gtrsim 1.94\,\chi_{\mathrm{rec}}\,. \tag{7}$$

With $\chi_{\mathrm{rec}} \simeq 3.15\,c/H_0$ in a Planck cosmology [12],

$$L \gtrsim 6\,\frac{c}{H_0} \tag{8}$$

as an order-of-magnitude statement (exact numerical factors depend on the statistic and priors). Recent topology surveys likewise place the shortest E_1 side near the diameter scale of the last-scattering surface, not one Hubble radius [6, 8].

Write $L = \alpha\,c/H_0$. Empirical scale matching uses $\alpha = 1$; cubic bounds require $\alpha \sim \mathcal{O}(6)$. If the phenomenological formula is reinterpreted as $a_0 = c^2/(2\pi L) = cH_0/(2\pi\alpha)$, topology-safe α yields

$$a_0 \lesssim \frac{1}{6} \times \frac{cH_0}{2\pi} \sim 1.8 \times 10^{-11}\,\mathrm{m\,s^{-2}}\,, \tag{9}$$

which no longer reproduces the intended MOND-scale match. Therefore the cubic-E_1, $\alpha = 1$ package is incompatible with simultaneously targeting $a_0 \sim 10^{-10}$ and CMB non-detection.

Vague statements that a "Hubble-scale domain is compatible with non-detection" are not supported for cubic E_1 under these bounds.

V. PRESENT-EPOCH MATCHING VERSUS FIXED-COMOVING FLRW

For fixed comoving moduli $L^{(\mathrm{com})}$,

$$L_{\mathrm{phys}}(t) = a(t)\,L^{(\mathrm{com})}\,, \qquad \frac{\dot{L}_{\mathrm{phys}}}{L_{\mathrm{phys}}} = H\,. \tag{10}$$

Requiring instead $L_{\mathrm{phys}}(t) = c/H(t)$ gives

$$\frac{\dot{L}_{\mathrm{phys}}}{L_{\mathrm{phys}}} = -\frac{\dot{H}}{H}\,, \tag{11}$$

and equating the two yields $\dot{H} = -H^2$, i.e. deceleration parameter $q = 0$ (coasting). Exact tracking is therefore incompatible with ordinary matter-, radiation-, or Λ-dominated FLRW evolution under fixed comoving moduli.

The remaining alternatives are: (i) present-epoch numerical identification only; or (ii) an independent modulus sector with its own kinetic term and stress-energy. No such sector is constructed here.

The Hubble sphere c/H is also not generally a particle or event horizon; calling it a "global causal domain" therefore requires an additional causal definition [13].

VI. THE PROJECTOR RATIO DOES NOT DETERMINE $C_{\rm obs}$

A. The identity

For any n^i with $\gamma_{ij}n^in^j = 1$, define the metric trace by $\mathrm{Tr}_\gamma(h) = \gamma^{ij}h_{ij}$. Then

$$h_{ij} = \gamma_{ij} - n_i n_j\,, \qquad \frac{\mathrm{Tr}_\gamma(h)}{\mathrm{Tr}_\gamma(\gamma)} = \frac{2}{3}\,. \tag{12}$$

This is correct, topology-independent, and independent of superfluidity. It discards tensorial orientation information and is not special to T^3.

B. Static AQUAL-type action

A static AQUAL-type force sector [7] may be written

$$S_{\rm WF} = \int dt\, d^3x \left[-\frac{|\nabla\Phi_N|^2}{8\pi G} - \rho_b\Phi_N - \frac{C_{\rm IR}}{12\pi G\, a_0}\,|\nabla\psi|^3 - C_m\,\rho_b\,\psi \right], \tag{13}$$

Here $a_0 > 0$, $C_m > 0$, and $C_{\rm IR} > 0$ define the attractive positive-coefficient branch used below. The potentials have dimensions $[\Phi_N] = [\psi] = \mathrm{L}^2\mathrm{T}^{-2}$, so $[|\nabla\psi|] = [a_0] = \mathrm{LT}^{-2}$, while C_m and $C_{\rm IR}$ are dimensionless. Write $q = |\nabla\psi| = \sqrt{\gamma^{ij}\partial_i\psi\partial_j\psi}$. Neither n^i nor h_{ij} appears. The kinetic operator is fully three-dimensional and isotropic. There is therefore no field-theoretic sense in which Eq. (12) "matches" onto the EFT.

Variation yields

$$\nabla\cdot\left(\frac{|\nabla\psi|}{a_0}\,\nabla\psi\right) = 4\pi G\,\frac{C_m}{C_{\rm IR}}\,\rho_b\,. \tag{14}$$

In spherical symmetry, with $g_N = |\nabla\Phi_N|$ and physical acceleration $g_P = C_m|\nabla\psi|$,

$$g_P = C_{\rm obs}\sqrt{a_0\, g_N}\,, \qquad \boxed{C_{\rm obs} = \frac{C_m^{3/2}}{\sqrt{C_{\rm IR}}}}\,. \tag{15}$$

C. Field-rescaling invariance

Under $\psi \mapsto s\psi$ with $s > 0$,

$$C_m \mapsto \frac{C_m}{s}\,, \qquad C_{\rm IR} \mapsto \frac{C_{\rm IR}}{s^3}\,, \tag{16}$$

while $C_{\rm obs}$ is invariant. More generally, an invariant monomial obeys $-a - 3b = 0$ for $C_m^a C_{\rm IR}^b$; the response in Eq. (15) is the positive square root of the corresponding basic ratio $C_m^3/C_{\rm IR}$. Setting $C_m = C_{\rm IR}$ so that $C_{\rm obs} = C$ is a normalization convention, not a calculation of $C = 2/3$.

a. Scope of the result. Equation (12) alone does not support a statement that $C_{\rm obs} = 2/3$ follows from T^3 geometry or from "matching." Such a derivation would require inserting h^{ij} into the action, specifying its dynamical origin, and integrating out the relevant fields.

D. Compact sources

On compact T^3, the integral of any pure divergence vanishes, so localized solutions must use a compensated density contrast (and, if present, a zero-mean exchange source). For a flat periodic metric, general morphology, and the local-static baseline with vanishing exchange source,

$$\frac{|\nabla\psi|}{a_0}\,\nabla\psi = \frac{C_m}{C_{\rm IR}}\,\nabla\Phi_N + \nabla\times\mathbf{H} + \mathbf{h}_0\,, \tag{17}$$

where $\mathbf{H}$ is periodic and $\mathbf{h}_0$ is the constant harmonic flux fixed by the solution. A periodic curl has zero spatial mean, but the nonlinear gradient flux need not, so dropping $\mathbf{h}_0$ needs an additional condition. It is not a freely fitted uniform acceleration. A nonzero exchange source requires its own source contribution before the difference is divergence-free; a curved metric requires the corresponding metric-dependent Hodge decomposition. Pointwise square-root formulae remain approximations for disks. Neither this decomposition nor compact topology derives $C_{\rm obs} = 2/3$.

VII. CONDITIONAL RAR NORMALIZATION CHECK IN THE SPHERICAL BASELINE

This section is deliberately conditional. It uses the static AQUAL-type action of Eq. (13), the positive-mass spherical branch, regular-origin flux, and no added homogeneous flux. It then imposes $C_{\rm obs} = 2/3$ as a comparator; the preceding section does not derive that value. The result is therefore neither a global periodic-T^3 solution nor a disk-galaxy fit.

In the deep-MOND regime, $g_P = C_{\rm obs}\sqrt{a_0 g_N}$ is equivalent to

$$g_P = \sqrt{a_{0,\rm eff}\, g_N}\,, \qquad a_{0,\rm eff} = C_{\rm obs}^2\, a_0\,. \tag{18}$$

Imposing both $a_0 = cH_0/2\pi \approx 1.1\times10^{-10}\,\mathrm{m\,s^{-2}}$ and $C_{\rm obs} = 2/3$ yields

$$a_{0,\rm eff} = \frac{4}{9}a_0 \approx 4.9\times10^{-11}\,\mathrm{m\,s^{-2}}\,, \tag{19}$$

whereas modern RAR determinations find $a_0 \simeq 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$ [3, 4], a factor ~ 2.4 normalization difference under these assumptions. Equivalently, asymptotic baryonic Tully–Fisher velocities scale as $v^4 \propto C_{\rm obs}^2 a_0 G M_b$; $C_{\rm obs} = 2/3$ lowers v by $\sim 18\%$ at fixed baryonic mass. That analytic shift is large enough to require an explicit

likelihood test, but it is not by itself a completed empirical rejection.

Within this branch, one cannot simultaneously identify $a_0 = cH_0/2\pi$ with the empirical MOND scale and impose a universal $C_{\rm obs} = 2/3$ without changing the effective normalization. Consistent options for a future fit are:

1. set $C_{\rm obs} = 1$ (up to the definition of a_0);
2. redefine the input scale so that $C_{\rm obs}^2 a_0$ matches the RAR;
3. drop universality and derive a morphology-dependent response from a solved disk PDE before fitting.

VIII. EVALUATION CRITERIA, RETAINED RESULTS, AND A RECONSTRUCTION PATH

A. What this note establishes

1. The coincidence (1) is empirical motivation only.
2. $a_0 \equiv cH_0/2\pi$ is allowed solely as a *present-epoch phenomenological postulate*.
3. The weak-field combination $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$ is a derived, normalization-invariant observable coefficient [Eq. (15)].
4. On compact T^3, sources must be compensated and disk solutions generally admit a curl term [Eq. (17)].
5. The scoped incompatibilities in Secs. III–VII constrain the tested cubic geometric packaging of those two numbers without deciding MOND or alternative microscopic constructions as a class.

B. Path required before any geometric-derivation claim

A subsequent work that claims to *derive* a_0 or $C_{\rm obs}$ from topology or microphysics must close the following chain. Items are grouped by sector, and each is a separately testable technical requirement.

a. Microphysics and circulation.

(M1) Define a condensate order parameter $\Psi = \sqrt{\rho}\, e^{i\theta}$ (or a relativistic analogue) with a definite mass parameter m.

(M2) Obtain the circulation from $\nabla\theta$ and impose integer winding on non-contractible cycles of T^3, yielding $\Gamma = n\,h/m$ rather than $\Gamma = cL$.

(M3) Fix a single geometric meaning of L (cube edge, closed geodesic length, or circumference) and use it consistently in every formula.

(M4) Derive a map $(\Gamma, m, n, L) \mapsto a_0$ that does not rely on a chart-dependent "distribution" of cH_0 over an angular range of 2π.

b. Cosmic scale and moduli.

(S1) Either restrict scale matching to the present epoch, or supply a modulus action for $L_{\rm phys}(t)$ that does not force the coasting condition $q = 0$.

(S2) Confront the resulting L with Planck/COMPACT bounds for cubic E_1 and for twisted flat alternatives [6, 8].

c. Weak-field coefficient and data.

(E1) If a projector h_{ij} is essential, insert it into the action, specify the dynamics of n^i, and compute $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$ as a definite number independent of field normalization.

(E2) Reconcile $C_{\rm obs}^2 a_0$ with the empirical RAR/BTFR normalization (avoid the factor-4/9 conflict of Sec. VII).

(E3) Solve the compensated periodic nonlinear equation for at least one spherical and one disk baryonic source, including any curl contribution.

d. Representation (already satisfied here).

(R1) Use a face-identified parallelepiped for T^3 (Fig. 1), never an embedded doughnut T^2.

Absent evidence closing (M1)–(M4), (S1)–(S2), and (E1)–(E3), this note supports present-epoch phenomenological matching of Eq. (1), the derived invariant $C_{\rm obs}$, and the scoped constraints on the tested cubic construction—not a geometric derivation of the MOND scale.

CODE AND DATA AVAILABILITY

Figure 1 is produced by `Scripts/itsm_t3_fundamental_domain.py`. Source code is available in the public GitHub repository and the project Zenodo archive. Scripts that render an embedded doughnut surface depict T^2 and should not be used as T^3 schematics. No new observational catalogue is analysed.

DECLARATION ON THE USE OF ARTIFICIAL-INTELLIGENCE TOOLS

Artificial-intelligence-assisted tools were used for editorial drafting, source-code assistance, and checks of symbolic and numerical consistency. The author independently reviewed the manuscript, equations, cited sources, and generated artifacts and accepts full responsibility for the content.

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